

Atomic-Unit Decomposition and Relation Inference

A Worked Example

FactReasoner — Ideation

June 16, 2026

1 Source paragraph

The defendant was found liable for breach of contract, which occurred on March 15, 2022, after they failed to deliver the specified goods. As a result, the court awarded the plaintiff \$50,000 in damages for the breach. However, the defendant argued that the plaintiff's own actions constituted a prior breach, thus excusing their performance. The case hinged on the initial agreement signed on January 1, 2022, which the court ultimately declared void ab initio (void from the beginning) due to fraudulent misrepresentation by the plaintiff. Separately, the court found the defendant liable for unjust enrichment and awarded the plaintiff \$10,000 on that basis, as the defendant had retained a down payment.

2 Atomic units (in order of appearance)

An *atomic unit* is a short, self-contained sentence carrying exactly one fact or claim.

- F1.** The defendant was found liable for breach of contract.
- F2.** The breach of contract occurred on March 15, 2022.
- F3.** The defendant failed to deliver the specified goods.
- F4.** The court awarded the plaintiff \$50,000 in damages for the breach.
- F5.** The defendant argued that the plaintiff's own actions constituted a prior breach.
- F6.** The defendant argued that the plaintiff's prior breach excused the defendant's performance.
- F7.** The case hinged on the initial agreement.
- F8.** The initial agreement was signed on January 1, 2022.
- F9.** The court declared the initial agreement void ab initio (void from the beginning).
- F10.** The agreement was declared void due to fraudulent misrepresentation by the plaintiff.
- F11.** The court found the defendant liable for unjust enrichment.
- F12.** The court awarded the plaintiff \$10,000 on the basis of unjust enrichment.
- F13.** The defendant had retained a down payment.

3 Relations

Logical prerequisites ($A \rightarrow B$)

- $F3 \rightarrow F1$ — failing to deliver the goods grounds the breach finding.
- $F1 \rightarrow F2$ — the breach must exist for it to have a date.
- $F1 \rightarrow F4$ — liability for breach is required to award damages “for the breach.”
- $F7 \rightarrow F1$ — an initial agreement must exist for there to be a contract to breach.
- $F7 \rightarrow F8$ — the agreement must exist for it to have a signing date.
- $F7 \rightarrow F9$ — the agreement must exist for the court to declare it void.
- $F8 \rightarrow F9$ — the agreement signed Jan 1, 2022 is the thing declared void.
- $F10 \rightarrow F9$ — fraudulent misrepresentation is the stated cause of the voidness.
- $F13 \rightarrow F11$ — retaining the down payment grounds the unjust-enrichment finding.
- $F11 \rightarrow F12$ — the unjust-enrichment finding grounds the \$10,000 award.

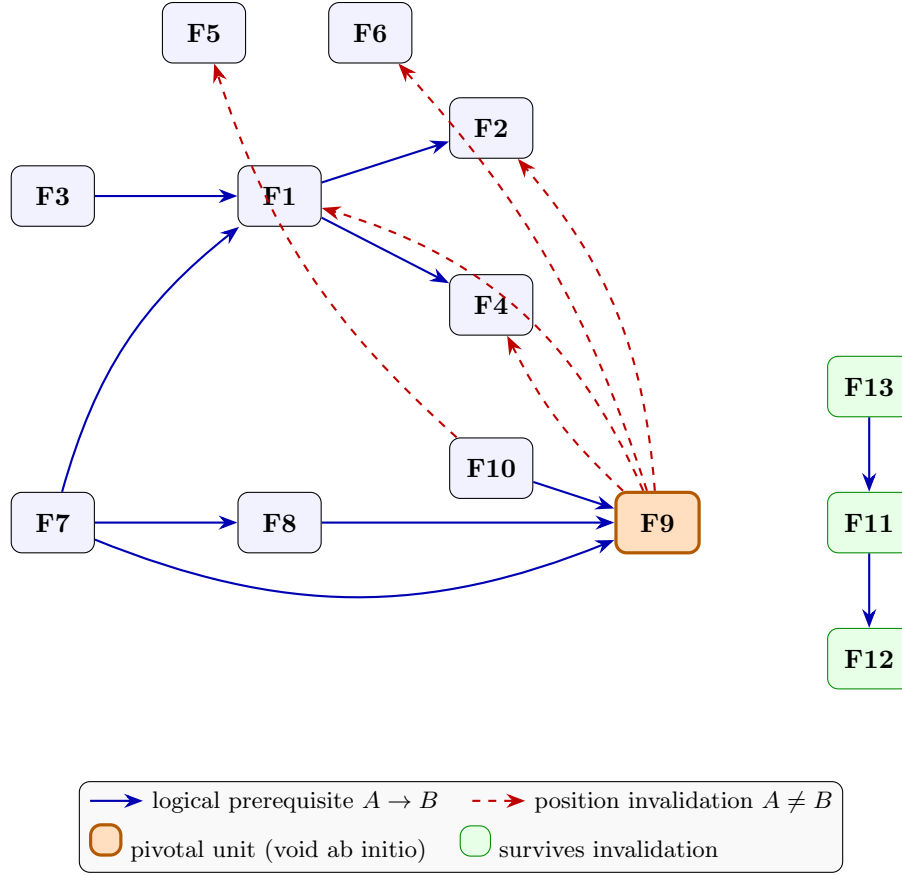
Position invalidations ($A \neq B$)

- $F9 \neq F1$ — void *ab initio* means no valid contract ever existed, so the earlier breach-of-contract finding is invalidated.
- $F9 \neq F2$ — a breach “on March 15, 2022” cannot stand once no valid contract existed on that date.
- $F9 \neq F4$ — the \$50,000 damages award rests on a breach that is now deemed never to have occurred.
- $F9 \neq F6$ — there was no valid performance obligation to be “excused.”
- $F10 \neq F5$ — the plaintiff’s fraud supersedes the defendant’s “prior breach by plaintiff” argument with a stronger ground.

Note. The unjust-enrichment chain (F11, F12, F13), flagged “Separately,” is *not* invalidated by F9: restitution for unjust enrichment applies precisely *because* no valid contract exists, so it is consistent with — even reinforced by — the void declaration.

4 Relation graph

Solid blue arrows are logical prerequisites ($A \rightarrow B$). Dashed red arrows are position invalidations ($A \neq B$). The pivotal unit **F9** (void *ab initio*) is highlighted: it retroactively invalidates the earlier breach claims, while the unjust-enrichment cluster on the right survives.



5 Second example: a biography

Source paragraph

Lanny Flaherty is an American actor born on December 18, 1949, in Pensacola, Florida. He has appeared in numerous films, television shows, and theater productions throughout his career, which began in the late 1970s. Some of his notable film credits include “King of New York,” “The Abyss,” “Natural Born Killers,” “The Game,” and “The Straight Story.” On television, he has appeared in shows such as “Law & Order,” “The Sopranos,” “Boardwalk Empire,” and “The Leftovers.” Flaherty has also worked extensively in theater, including productions at the Public Theater and the New York Shakespeare Festival. He is known for his distinctive looks and deep gravelly voice, which have made him a memorable character actor in the industry.

Atomic units (in order of appearance)

F1. Lanny Flaherty is an American actor.

F2. Lanny Flaherty was born on December 18, 1949.

F3. Lanny Flaherty was born in Pensacola, Florida.

F4. Lanny Flaherty has appeared in numerous films, television shows, and theater productions.

- F5.** Lanny Flaherty’s career began in the late 1970s.
- F6.** Lanny Flaherty’s notable film credits include “King of New York.”
- F7.** Lanny Flaherty’s notable film credits include “The Abyss.”
- F8.** Lanny Flaherty’s notable film credits include “Natural Born Killers.”
- F9.** Lanny Flaherty’s notable film credits include “The Game.”
- F10.** Lanny Flaherty’s notable film credits include “The Straight Story.”
- F11.** Lanny Flaherty has appeared in the television show “Law & Order.”
- F12.** Lanny Flaherty has appeared in the television show “The Sopranos.”
- F13.** Lanny Flaherty has appeared in the television show “Boardwalk Empire.”
- F14.** Lanny Flaherty has appeared in the television show “The Leftovers.”
- F15.** Lanny Flaherty has worked extensively in theater.
- F16.** Lanny Flaherty’s theater work includes productions at the Public Theater.
- F17.** Lanny Flaherty’s theater work includes productions at the New York Shakespeare Festival.
- F18.** Lanny Flaherty is known for his distinctive looks and deep gravelly voice.
- F19.** Lanny Flaherty’s distinctive looks and voice have made him a memorable character actor.

Relations

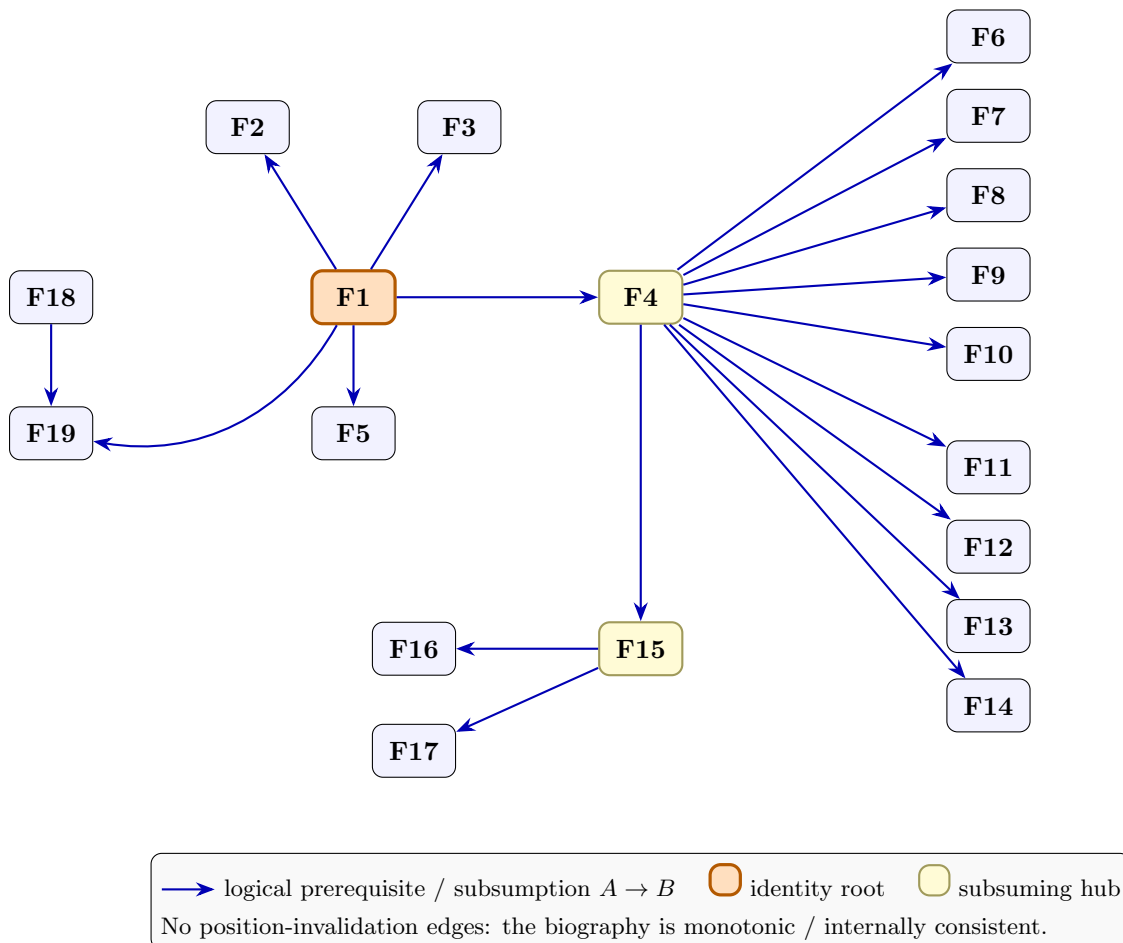
Logical prerequisites ($A \rightarrow B$). Here the prerequisite relation is mostly one of *subsumption* / *instantiation*: the general claim that he appeared in films, TV, and theater (F4) underpins each specific credit, and the identity claim (F1) underpins everything.

- $F1 \rightarrow F4$ — being an actor is presupposed by having a body of acting work.
- $F1 \rightarrow F5$ — being an actor is presupposed by having an acting career.
- $F4 \rightarrow F6, F7, F8, F9, F10$ — the general “appeared in films” claim subsumes each specific film credit.
- $F4 \rightarrow F11, F12, F13, F14$ — the general “appeared in TV shows” claim subsumes each specific show.
- $F4 \rightarrow F15$ — the general claim subsumes the theater-work claim.
- $F15 \rightarrow F16, F17$ — working in theater is presupposed by the specific theater venues.
- $F18 \rightarrow F19$ — the distinctive looks/voice are the stated cause of his being a memorable character actor.
- $F1 \rightarrow F19$ — being an actor is presupposed by being a memorable *character* actor.

Position invalidations ($A \neq B$). *None*. Unlike the legal example, this biography is monotonic: every later claim is consistent with the earlier ones, and no statement retracts or contradicts a prior position. The absence of invalidation edges is itself the finding.

Relation graph

All edges are logical prerequisites / subsumption ($A \rightarrow B$); there are no invalidation edges. The identity node **F1** is the root; the “body of work” node **F4** fans out to the specific credits.



A modified biography with planted contradictions

The biography above is internally consistent, so it has no invalidation edges. To illustrate the contradiction relation we now *alter* the passage, inserting later statements that conflict with earlier ones (different birth year and place, a different nationality, a different career-start decade, and a voice description that reverses the earlier one). Each contradiction is a positional invalidation: the earlier atom, asserted first, is contradicted by a later atom.

Modified source paragraph (contradictions in *italics*).

Lanny Flaherty is an American actor born on December 18, 1949, in Pensacola, Florida. His career began in the late 1970s, and he appeared in numerous films, television shows, and theater productions. He is known for his deep gravelly voice. *Born in 1942, Flaherty was a Mississippi native who first appeared on screen only in the 1990s.* Critics often described *the British character actor* for *his soft, high-pitched voice*, which made him instantly recognizable.

Atomic units (self-contained, in order of appearance).

- M1. Lanny Flaherty is an American actor.
- M2. Lanny Flaherty was born on December 18, 1949.
- M3. Lanny Flaherty was born in Pensacola, Florida.
- M4. Lanny Flaherty's career began in the late 1970s.
- M5. Lanny Flaherty appeared in numerous films, television shows, and theater productions.
- M6. Lanny Flaherty is known for his deep gravelly voice.
- M7. Lanny Flaherty was born in 1942.
- M8. Lanny Flaherty was a Mississippi native.
- M9. Lanny Flaherty first appeared on screen only in the 1990s.
- M10. Lanny Flaherty is a British character actor.
- M11. Lanny Flaherty has a soft, high-pitched voice.
- M12. Lanny Flaherty's voice made him instantly recognizable.

Relations

Logical prerequisites ($A \rightarrow B$).

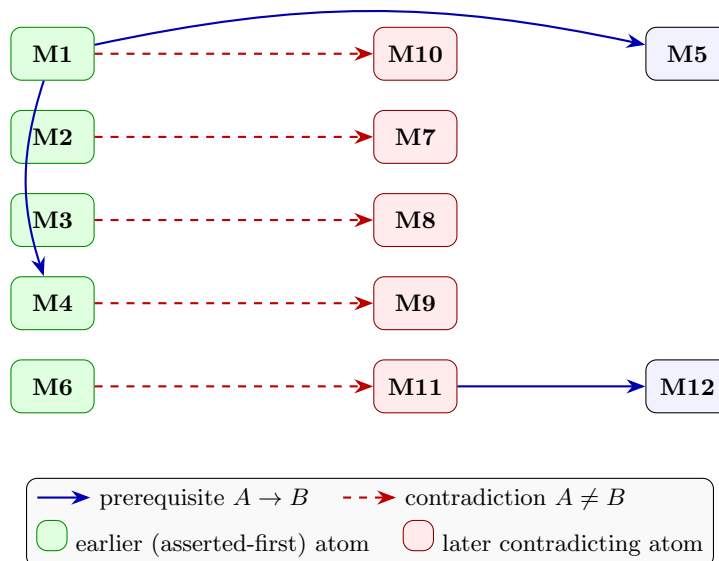
- $M1 \rightarrow M5$ — being an actor is presupposed by having a body of acting work.
- $M1 \rightarrow M4$ — being an actor is presupposed by having an acting career.
- $M11 \rightarrow M12$ — having a (distinctive) voice is presupposed by that voice making him recognizable.

Position invalidations ($A \neq B$). Each later atom contradicts an earlier one asserted at a different position.

- $M2 \neq M7$ — born December 18, 1949 contradicts born in 1942.
- $M3 \neq M8$ — born in Pensacola, Florida contradicts being a Mississippi native.
- $M4 \neq M9$ — a career beginning in the late 1970s contradicts first appearing on screen only in the 1990s.
- $M1 \neq M10$ — being an *American* actor contradicts being a *British* character actor.
- $M6 \neq M11$ — a deep gravelly voice contradicts a soft, high-pitched voice.

Note. The contradictions are pairwise and local: each invalidation links exactly one early biographical claim to its later mutually exclusive restatement. The prerequisite structure (identity $\rightarrow$ work / career, voice $\rightarrow$ recognizability) is unaffected; M5 has no specific-credit children here because the modified paragraph drops the individual titles.

Relation graph. Solid blue arrows are logical prerequisites ($A \rightarrow B$). Dashed red arrows are position invalidations ($A \neq B$): each links an earlier (*green, asserted-first*) atom to the later (*red*) atom that contradicts it. The five contradiction pairs are stacked so no arrow crosses a node.



6 Third example: a narrative passage

Source paragraph

Elinor is surprised no letter arrives with details about the marriage, and she wonders how Edward felt being close to Barton. Elinor asks her mother to write to Colonel Brandon for news, realizing she hoped something would prevent Edward's marriage. Elinor feels incredibly hurt by Edward's marriage. Colonel Brandon arrives at the house, but it is revealed to be Edward instead. Elinor is surprised that Edward and Lucy were married so soon, and she mulls over the event of Edward's marriage, supposing that the Ferrars must be settled at Delaford, envisioning Edward and Lucy's life there. Edward enters the room looking ill with anxiety, and Elinor tells herself to be calm upon Edward's arrival. Elinor thinks someone in London might have informed her about Edward and Lucy's marriage. Marianne and Margaret retreat, leaving Elinor and Mrs. Dashwood with Edward, while Marianne and Mrs. Dashwood react with shock and uncertainty about Edward's arrival. Mrs. Dashwood breaks the ice by shaking Edward's hand and wishing him happiness. However, no word has arrived from Colonel Brandon, and Mrs. Dashwood asks about Mrs. Ferrars' health, to which Elinor inquires if Mrs. Ferrars is at Longstaple. Edward is shocked to discover his mother is not nearby, and Elinor pointedly asks about Mrs. Edward Ferrars. Edward is confused and asks if Elinor means Mrs. Robert Ferrars instead, leading to everyone being shocked by the revelation that Robert married Lucy Steele. Elinor flees the room, unsure of how to react, and bursts into tears of joy. Edward sits in silence, unsure of what to do, then simply leaves the room. Everyone is left perplexed by the situation.

Atomic units (self-contained, in order of appearance)

Pronouns are resolved to their referents so each unit stands alone.

- F1.** Elinor is surprised that no letter arrives with details about the marriage.
- F2.** Elinor wonders how Edward felt being close to Barton.
- F3.** Elinor asks her mother to write to Colonel Brandon for news.
- F4.** Elinor realizes she had hoped something would prevent Edward's marriage.
- F5.** Elinor feels incredibly hurt by Edward's marriage.
- F6.** A visitor arrives at the house and is initially taken to be Colonel Brandon.
- F7.** The visitor is revealed to be Edward instead of Colonel Brandon.
- F8.** Elinor is surprised that Edward and Lucy were married so soon.
- F9.** Elinor mulls over the event of Edward's marriage.
- F10.** Elinor supposes that the Ferrars must be settled at Delaford.
- F11.** Elinor envisions Edward and Lucy's life at Delaford.
- F12.** Edward enters the room looking ill with anxiety.
- F13.** Elinor tells herself to be calm upon Edward's arrival.
- F14.** Elinor thinks someone in London might have informed her about Edward and Lucy's marriage.
- F15.** Marianne and Margaret retreat from the room.
- F16.** Marianne and Margaret leave Elinor and Mrs. Dashwood alone with Edward.
- F17.** Marianne and Mrs. Dashwood react with shock and uncertainty about Edward's arrival.
- F18.** Mrs. Dashwood breaks the ice by shaking Edward's hand.
- F19.** Mrs. Dashwood wishes Edward happiness.
- F20.** No word has arrived from Colonel Brandon.
- F21.** Mrs. Dashwood asks about Mrs. Ferrars' health.
- F22.** Elinor inquires whether Mrs. Ferrars is at Longstaple.
- F23.** Edward is shocked to discover Edward's mother is not nearby.
- F24.** Elinor pointedly asks about Mrs. Edward Ferrars.
- F25.** Edward is confused by Elinor's question about Mrs. Edward Ferrars.
- F26.** Edward asks whether Elinor means Mrs. Robert Ferrars instead.
- F27.** It is revealed that Robert married Lucy Steele.
- F28.** Everyone is shocked by the revelation that Robert married Lucy Steele.
- F29.** Elinor flees the room, unsure of how to react.

F30. Elinor bursts into tears of joy.

F31. Edward sits in silence, unsure of what to do.

F32. Edward then leaves the room.

F33. Everyone is left perplexed by the situation.

Relations

Logical prerequisites ($A \rightarrow B$). In a narrative the prerequisite relation is largely *enabling causation* / *temporal grounding*: an earlier event makes a later one possible or meaningful.

- $F6 \rightarrow F7$ — a visitor must arrive before being re-identified as Edward.
- $F7 \rightarrow F12$ — Edward being the arrival is presupposed by Edward entering the room.
- $F9 \rightarrow F10$ — mulling over the marriage grounds the supposition about Delaford.
- $F10 \rightarrow F11$ — supposing the Ferrars settle at Delaford grounds envisioning their life there.
- $F15 \rightarrow F16$ — Marianne and Margaret must retreat for Elinor and Mrs. Dashwood to be left with Edward.
- $F22 \rightarrow F23$ — Elinor’s question about Longstaple prompts Edward’s discovery his mother is not nearby.
- $F24 \rightarrow F25$ — the pointed question about “Mrs. Edward Ferrars” is what confuses Edward.
- $F25 \rightarrow F26$ — Edward’s confusion grounds his counter-question about Mrs. Robert Ferrars.
- $F26 \rightarrow F27$ — Edward’s counter-question surfaces the revelation that Robert married Lucy.
- $F27 \rightarrow F28$ — the revelation is what shocks everyone.
- $F27 \rightarrow F30$ — the revelation (Edward is *not* married) is why Elinor’s tears are tears of *joy*.

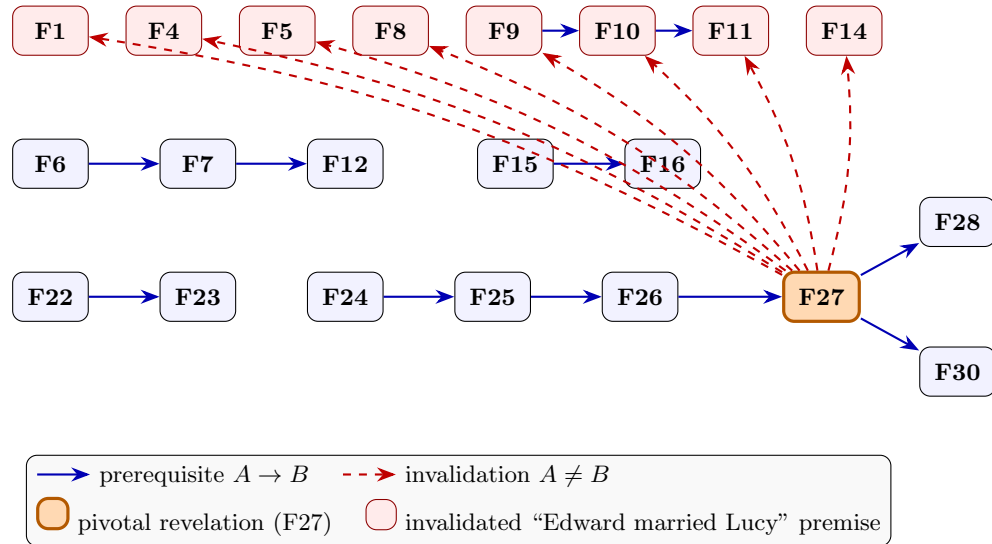
Position invalidations ($A \neq B$). The pivotal unit is **F27**: the revelation that it was *Robert* (not Edward) who married Lucy Steele. Appearing late in the passage, it retroactively invalidates the false premise – woven through the entire opening – that *Edward* married Lucy.

- $F27 \neq F1$ — the awaited “marriage” news concerned a marriage of Edward that never happened.
- $F27 \neq F4$ — there was no marriage of Edward for Elinor to have hoped to prevent.
- $F27 \neq F5$ — Elinor’s hurt rested on a marriage of Edward that did not occur.
- $F27 \neq F8$ — “Edward and Lucy were married so soon” is false; Edward did not marry Lucy.
- $F27 \neq F9$ — there was no “event of Edward’s marriage” to mull over.
- $F27 \neq F10$ — the Ferrars settling at Delaford as a married couple (Edward + Lucy) is invalidated.
- $F27 \neq F11$ — the envisioned life of Edward and Lucy at Delaford never existed.
- $F27 \neq F14$ — the supposed London report of *Edward and Lucy’s* marriage was mistaken.

Note. The invalidations all share a single source (F27) and a single target premise (“Edward married Lucy”). The events of arrival, greeting, and dialogue (F6, F7, F12–F26) are *not* invalidated – they actually happened; only Elinor’s (and the narrator’s) interpretation of *who* was married is overturned.

Relation graph

Solid blue arrows are logical prerequisites ($A \rightarrow B$). Dashed red arrows are position invalidations ($A \neq B$), all emanating from the pivotal revelation **F27**. The eight invalidated “Edward-married-Lucy” premises are tinted red; F27 is highlighted.



(For legibility the graph shows the principal chains and the full invalidation fan; purely sequential narration edges such as F1–F2 or F31–F32 are omitted. The mapping of these units onto the ground-truth scene below is given in Table 1, and the ground-truth dependency graph in Figure 1.)

Comparison against the ground-truth passage

The paragraph analysed above is a paraphrase of the following *ground-truth* account of the same scene. We re-decompose the ground truth, align it to the units F1–F33, and check that the relation structure (the pivot at the “Robert married Lucy” revelation, and the invalidation fan it triggers) is preserved.

Elinor finds herself mulling over the event of Edward’s marriage — she realizes that she’d hoped that something would come up to prevent it, but now that it’s happened, she feels incredibly hurt. She’s surprised that he and Lucy were married so soon, before he could possibly have been ordained. She wonders how he must have felt, being so close to Barton. She supposes that the Ferrars must be settled at Delaford already, and envisions their life there. Elinor had thought that someone in London might have seen fit to tell her of Edward and Lucy’s marriage, and she’s surprised when no letter arrives with details. She even asks her mother to write to Colonel Brandon, eager for any news. However, no word has arrived. As soon as she asks this, though, Colonel Brandon himself arrives — but wait, it’s not the Colonel. In fact, it’s Edward! Elinor tells herself to be calm. Marianne and Mrs. Dashwood both freak out as well, unsure

of what they should do. They silently await his arrival. Edward enters, looking ill with anxiety. Mrs. Dashwood breaks the ice by shaking his hand and wishing him happiness. Marianne and Margaret both retreat, leaving Elinor and Mrs. Dashwood to deal with the visitor. Mrs. Dashwood asks if Mrs. Ferrars is well, and Elinor, following up, asks if she's at Longstaple. Edward is shocked — his mother isn't anywhere near here! Elinor pointedly asks about Mrs. Edward Ferrars. All eyes turn to Edward. Edward stammers, confused — does Elinor mean Mrs. Robert Ferrars instead? Everyone is shocked: what's the meaning of this? This is the deal — it turns out that Robert, Edward's younger brother, just married Lucy Steele. Whoa. Our minds are blown. Elinor isn't sure how to react, so she flees the room and bursts into tears of joy. Edward also doesn't know what to do, so he sits in silence for a moment, then simply leaves. Everyone is totally perplexed.

Ground-truth atomic units (self-contained, in order of appearance).

- G1.** Elinor mulls over the event of Edward's marriage.
- G2.** Elinor realizes she had hoped something would come up to prevent Edward's marriage.
- G3.** Elinor feels incredibly hurt now that Edward's marriage has happened.
- G4.** Elinor is surprised that Edward and Lucy were married so soon.
- G5.** Edward and Lucy were married before Edward could possibly have been ordained.
- G6.** Elinor wonders how Edward must have felt being so close to Barton.
- G7.** Elinor supposes that the Ferrars must be settled at Delaford already.
- G8.** Elinor envisions the Ferrars' life at Delaford.
- G9.** Elinor had thought someone in London might tell her of Edward and Lucy's marriage.
- G10.** Elinor is surprised when no letter arrives with details of the marriage.
- G11.** Elinor asks her mother to write to Colonel Brandon for news.
- G12.** No word has arrived from Colonel Brandon.
- G13.** As soon as Elinor asks, a visitor arrives and is taken to be Colonel Brandon.
- G14.** The visitor is in fact Edward, not Colonel Brandon.
- G15.** Elinor tells herself to be calm.
- G16.** Marianne and Mrs. Dashwood are alarmed and unsure what to do.
- G17.** Marianne and Mrs. Dashwood silently await Edward's arrival.
- G18.** Edward enters the room looking ill with anxiety.
- G19.** Mrs. Dashwood breaks the ice by shaking Edward's hand and wishing him happiness.
- G20.** Marianne and Margaret retreat, leaving Elinor and Mrs. Dashwood with the visitor.

- G21.** Mrs. Dashwood asks whether Mrs. Ferrars is well.
- G22.** Elinor asks whether Mrs. Ferrars is at Longstaple.
- G23.** Edward is shocked because Edward’s mother is not anywhere near Barton.
- G24.** Elinor pointedly asks about Mrs. Edward Ferrars.
- G25.** All eyes turn to Edward.
- G26.** Edward, confused, asks whether Elinor means Mrs. Robert Ferrars instead.
- G27.** Everyone is shocked and asks the meaning of this.
- G28.** Robert, Edward’s younger brother, just married Lucy Steele.
- G29.** Elinor, unsure how to react, flees the room and bursts into tears of joy.
- G30.** Edward, unsure what to do, sits in silence and then leaves.
- G31.** Everyone is totally perplexed.

Alignment to F1–F33. Table 1 maps each predicted unit to its ground-truth counterpart. The two decompositions agree on the scene’s content; the ground truth additionally makes explicit two facts the paraphrase dropped — that the marriage preceded Edward’s ordination (**G5**) and that Robert is *Edward’s younger brother* (sharpening **G28** vs. predicted F27).

Ground-truth dependency graph. Figure 1 draws the relations on the ground-truth units, using the same conventions as the predicted graph above. The structure is preserved: **G28** (Robert married Lucy) is the pivot, and it invalidates exactly the units that presuppose “Edward married Lucy” (G1–G4, G7–G10) — the same premise set as the predicted F27 fan. The extra ground-truth fact **G5** (“married before ordination”) is likewise invalidated, since the marriage it describes is Edward’s.

Temporal-order discrepancies

The ground-truth units **G1–G31** are listed in **chronological order**: G_i happens before G_j whenever $i < j$. The altered passage, however, narrates the *same* events in a different order. Using the alignment of Table 1, we assign each predicted unit its *chronological rank* (the index of the ground-truth event it reports) and ask whether reading F1, F2, . . . , F33 in order visits those ranks in increasing order. Wherever a later predicted unit reports an *earlier* event than one before it, the narration has jumped backwards in time — a temporal discrepancy.

Chronological rank of each predicted unit.

F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
rank (G)	10	6	11	2	3	13	14	4	1	7	8	18	15	9	20	20	16
F	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	
rank (G)	19	19	12	21	22	23	24	25	26	28	27	29	29	30	30	31	

In a perfectly ordered paraphrase this rank sequence would be non-decreasing. Instead it lurches: it opens at rank 10, drops to rank 2 (F4), reaches back to rank **1** at F9 (the chronological *start* of the scene, narrated ninth), and only settles into order from F21 onward. Two discrepancies survive into the otherwise-ordered tail: F20 (rank 12, the missing word from Brandon, displaced very late) and the climactic local swap F27/F28 (rank 28 before rank 27).

Predicted	Ground truth	Note
F1, F14	G10, G9	awaited / missing news of the marriage
F4	G2	hoped to prevent it
F5	G3	feels hurt
F8	G4	surprised married so soon
F9	G1	mullets over the marriage
F10, F11	G7, G8	Delaforde supposition / envisioned life
F2	G6	wonders about Barton
F3	G11	asks mother to write to Brandon
F20	G12	no word from Brandon
F6, F7	G13, G14	“Brandon” arrives → is Edward
F12	G18	Edward enters, ill with anxiety
F13	G15	Elinor tells herself to be calm
F17	G16, G17	others alarmed; silently await
F18, F19	G19	Mrs. Dashwood greets / wishes happiness
F15, F16	G20	Marianne & Margaret retreat
F21	G21	asks if Mrs. Ferrars is well
F22	G22	asks if at Longstaple
F23	G23	Edward shocked: mother not near
F24	G24	pointed “Mrs. Edward Ferrars”
F25, F26	G26 (+G25)	confused; “Mrs. Robert Ferrars?”
F27	G28	<i>pivot</i> : Robert (Edward’s brother) married Lucy
F28	G27	everyone shocked
F29, F30	G29	Elinor flees; tears of joy
F31, F32	G30	Edward silent, then leaves
F33	G31	everyone perplexed
–	G5	<i>ground-truth only</i> : married before ordination

Table 1: Alignment of the predicted units F1–F33 to the ground-truth units G1–G31. Bold marks the shared pivot.

What is out of place.

- **Scrambled opening cluster (F1–F14).** Chronologically the scene runs G1 (mullets)→G2 (hoped to prevent)→G3 (hurt)→ G4 (so soon)→G6 (Barton)→G7/8 (Delaforde)→ G9/10 (London / no letter)→G11 (asks mother)→G12 (no word). The paraphrase instead *opens* with the missing-letter surprise (F1=G10) and the Barton musing (F2=G6), asks the mother (F3=G11) *before* establishing the hurt (F4/F5=G2/G3), and reaches the true starting beat “Elinor mullets over the marriage” (F9=G1) only ninth.
- **Displaced “no word from Brandon” (F20=G12).** In the ground truth this immediately follows the request to write (G11); the paraphrase moves it past the arrival and greeting, to just before Mrs. Dashwood’s question about Mrs. Ferrars.
- **Climactic swap (F27↔F28, i.e. G28 before G27).** The paraphrase states the revelation (Robert married Lucy) and then reports the general shock; the ground truth has the shocked “what’s the meaning of this?” (G27) precede the explanatory reveal (G28).

Discrepancy plot. Figure 2 plots each predicted unit at (appearance order, chronological rank). A faithful retelling would lie on the rising dotted diagonal; every point *below* that diagonal is an event told later than its true time, and every **red back-arc** connects consecutive units where

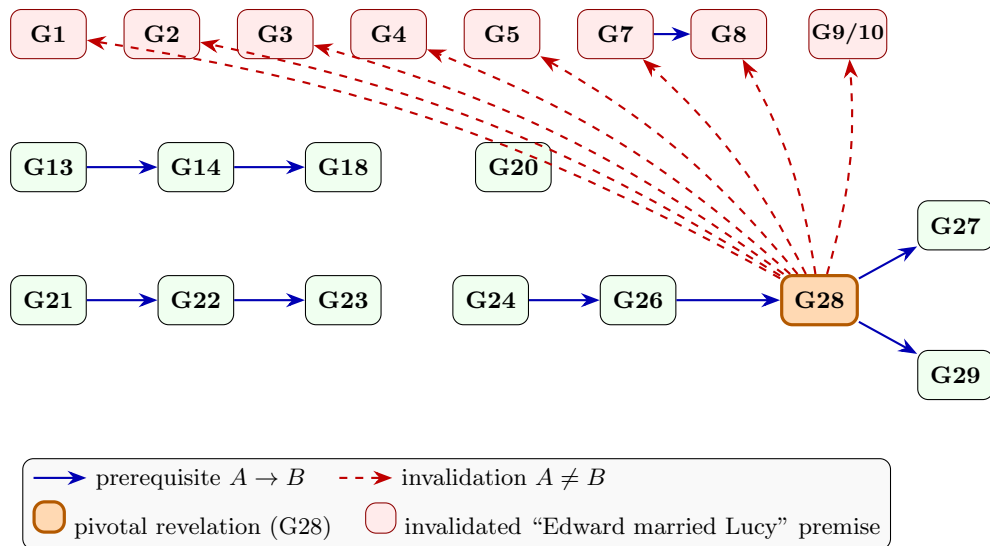


Figure 1: Ground-truth dependency graph (units G1–G31). The pivot **G28** mirrors the predicted pivot F27 (Table 1); its invalidation fan covers the same premise set, plus the ground-truth-only unit G5 (married before ordination).

the story jumps backwards in time. The dense tangle of back-arcs over F1–F14 is the scrambled opening; the lone late arcs are F20 and the F27/F28 swap.

7 Fourth example: synthesizing a summary from a reliable and an unreliable source

This example starts from two source paragraphs about the same subject. Source **A** is largely accurate; source **B** is a sensational news piece containing several factual errors that *contradict* A. We then synthesize a summary **S** that deliberately interleaves claims from both, so that S contains both logical-implication chains and positional invalidations, and we analyse S with the usual decomposition.

Source A (reliable)

Keyara Jones is an American college basketball player who played in the NCAA Division I women’s basketball league. Jones played college basketball for the University of Alabama during the 2020–2021 season, as a member of the Crimson Tide women’s basketball team, which competes in the Southeastern Conference (SEC) of NCAA Division I. The NCAA Division I women’s basketball league consists of 356 teams and is considered the highest level of women’s college basketball in the United States. According to the University of Alabama’s athletic website, Jones is listed as a *guard from Birmingham, Alabama*, and was a member of the women’s basketball team during the 2020–2021 season; ESPN and CBS Sports also reported on her performances that season.

Source B (unreliable — contains factual errors)

“Breaking News: Alabama’s Keyara Jones Makes History as First Female Player in NCAA Division I *Men’s* Basketball.” The 20-year-old junior, a 5’10” *guard from Montgomery, Alabama*, made her debut last night. Averaging 18 points and 7 rebounds per game, she led the Alabama

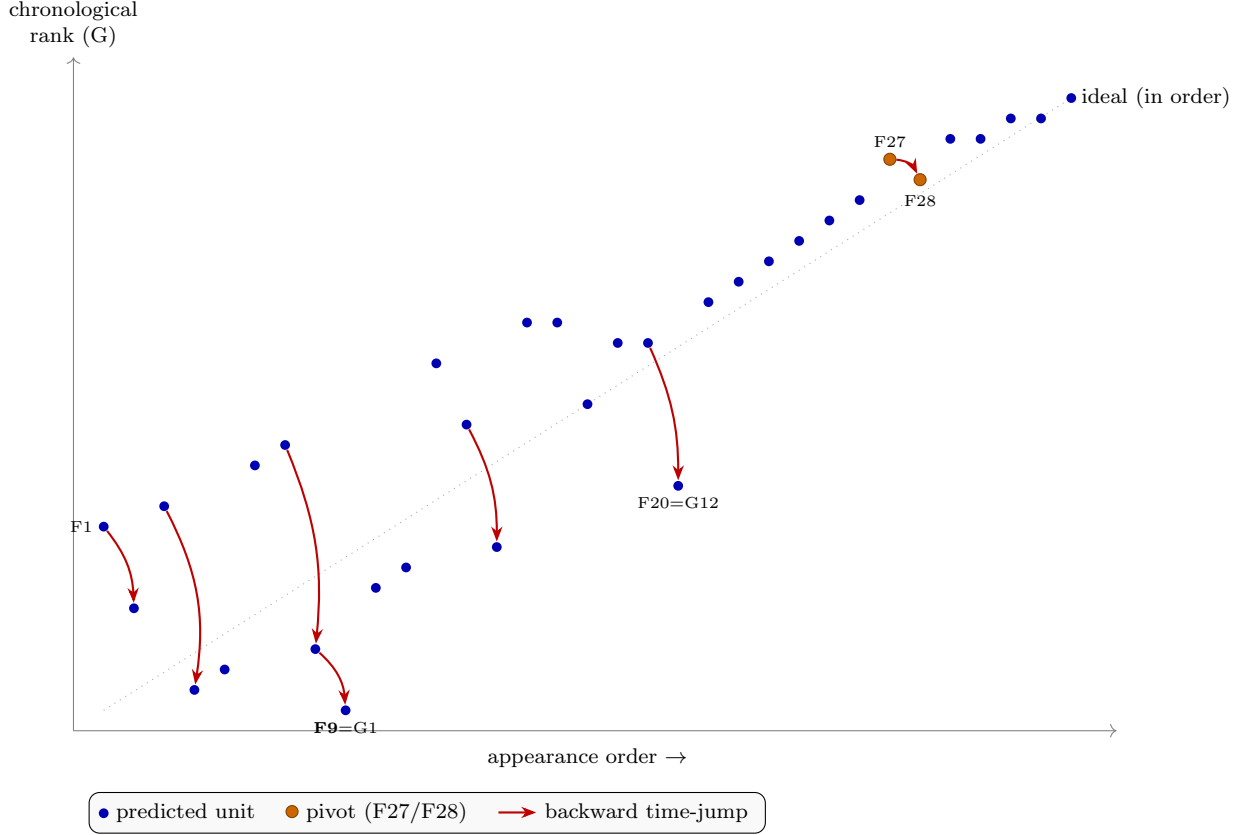


Figure 2: Temporal-order discrepancy plot. Each predicted unit F_i is placed at its appearance order (x) and the chronological rank of the ground-truth event it reports (y). A faithful retelling tracks the dotted diagonal; points below it are told late, and red arcs mark consecutive backward jumps in time. The opening cluster F_1 – F_{14} is badly scrambled (it reaches the true first event, $F_9=G_1$, only ninth), while F_{21} – F_{33} is ordered apart from the displaced F_{20} and the climactic F_{27}/F_{28} swap.

women’s team to a Sweet 16 appearance last season. Coach Nate Oats praised her, and a sports sociologist called it a milestone for gender equality. “According to NCAA data, only 12% of Division I men’s basketball players are women,” the piece reports, framing Jones’ entry as a step toward greater gender representation; several NBA teams are said to be scouting her.

Errors in B (each contradicts A). (i) Jones joined an NCAA Division I *men’s* team — A has her on the *women’s* team. (ii) She is a guard from *Montgomery* — A’s cited source says *Birmingham*. (iii) The “12% of men’s players are women” statistic is internally absurd. The fabricated junior/height/Sweet 16/NBA-scouting details have no support in A.

Synthesized summary S

S is written to interleave B’s sensational (false) framing *first* with A’s reliable record *afterward*, so that the later (true) atoms positionally invalidate the earlier (false) ones, while several implication chains run within each block.

Keyara Jones is an American college basketball player from Montgomery, Alabama, who plays guard for the University of Alabama. During the 2020–2021 season she made

history as the first woman to join an NCAA Division I men's basketball team, averaging 18 points per game and leading Alabama to a Sweet 16 appearance. Because she competed in the men's league, her achievement was hailed as a milestone for gender equality, and NCAA data showing that 12% of Division I men's players are women was cited to underscore the trend. In fact, the University of Alabama's athletic website lists Jones as a guard from Birmingham, Alabama, who was a member of the women's basketball team during the 2020–2021 season. The Crimson Tide women's team competes in the Southeastern Conference of NCAA Division I women's basketball, which is the highest level of women's college basketball in the United States and consists of 356 teams.

Atomic units of S (self-contained, in order of appearance)

- S1.** Keyara Jones is an American college basketball player.
- S2.** Keyara Jones is from Montgomery, Alabama.
- S3.** Keyara Jones plays guard for the University of Alabama.
- S4.** During the 2020–2021 season Keyara Jones made history as the first woman to join an NCAA Division I men's basketball team.
- S5.** Keyara Jones averaged 18 points per game.
- S6.** Keyara Jones led Alabama to a Sweet 16 appearance.
- S7.** Keyara Jones competed in the men's basketball league.
- S8.** Keyara Jones' achievement was hailed as a milestone for gender equality.
- S9.** NCAA data show that 12% of Division I men's basketball players are women.
- S10.** The 12% statistic was cited to underscore the gender-equality trend.
- S11.** The University of Alabama's athletic website lists Keyara Jones as a guard from Birmingham, Alabama.
- S12.** Keyara Jones was a member of the University of Alabama women's basketball team during the 2020–2021 season.
- S13.** The Crimson Tide women's team competes in the Southeastern Conference of NCAA Division I women's basketball.
- S14.** NCAA Division I women's basketball is the highest level of women's college basketball in the United States.
- S15.** NCAA Division I women's basketball consists of 356 teams.

Relations

Logical prerequisites ($A \rightarrow B$).

- $S3 \rightarrow S4$ — playing for Alabama is presupposed by joining an Alabama (men's) team.
- $S4 \rightarrow S7$ — joining a men's team is what it means to compete in the men's league.

- $S7 \rightarrow S8$ — competing in the men’s league is the stated ground for the gender-equality acclaim.
- $S9 \rightarrow S10$ — the 12% statistic is what is cited to underscore the trend.
- $S8, S10$ share the gender-equality framing ($S10$ reinforces $S8$).
- $S12 \rightarrow S13$ — membership of the women’s team is presupposed by that team competing in the SEC.
- $S13 \rightarrow S14$ — the women’s league existing/being top-level is presupposed by $S13$ placing the team in it.
- $S14 \rightarrow S15$ — the league being characterised (top level) is presupposed by counting its 356 teams.
- $S1 \rightarrow S3$ — being a college basketball player is presupposed by playing guard for a specific program.

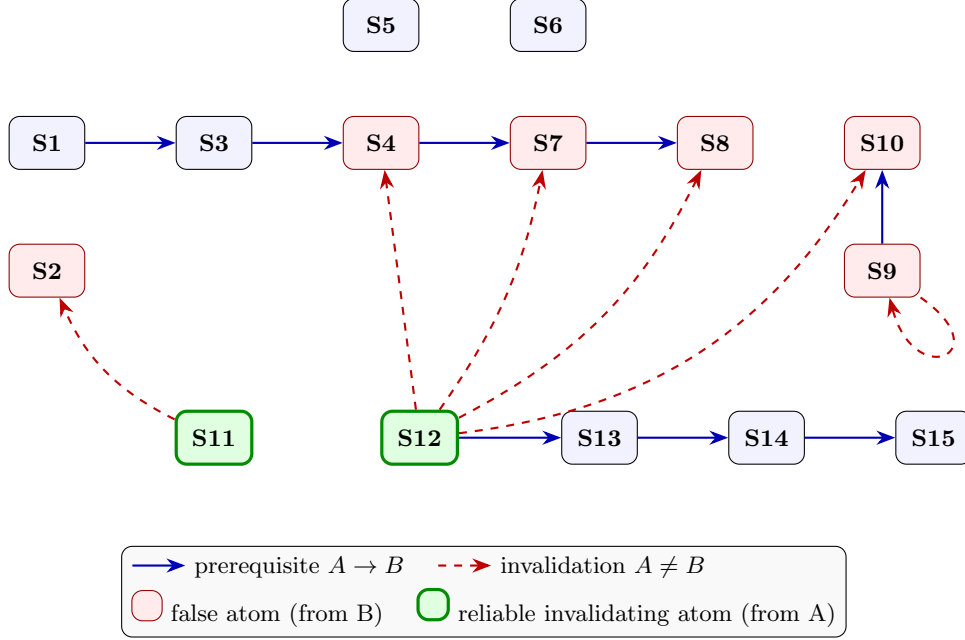
Position invalidations ($A \neq B$). The reliable atoms $S11$ and $S12$ (from source A), appearing *after* B’s framing, positionally invalidate the earlier false atoms.

- $S11 \neq S2$ — the cited website says *Birmingham*, contradicting “from Montgomery.”
- $S12 \neq S4$ — being on the *women’s* team contradicts “first woman on a men’s team.”
- $S12 \neq S7$ — women’s-team membership contradicts “competed in the men’s league.”
- $S12 \neq S8$ — with no men’s-league entry, the gender-equality milestone collapses.
- $S12 \neq S10$ — the men’s-representation framing is voided once the men’s-league premise fails.
- $S9 \neq S9$ (self-inconsistent) — “12% of *men’s* players are women” is internally contradictory regardless of position.

Note. The Sweet 16 and 18-ppg atoms ($S5, S6$) are unsupported by A but not directly contradicted by it; they are left as neither implied nor invalidated. The contradiction structure is concentrated on $S12$ (true women’s-team membership) overturning the men’s-league thread $S4$ – $S10$.

Relation graph

Solid blue arrows are logical prerequisites ($A \rightarrow B$); dashed red arrows are position invalidations ($A \neq B$). False atoms imported from B are tinted red; the two reliable invalidating atoms from A ($S11, S12$) are green and highlighted. The self-contradictory statistic $S9$ carries a red self-loop.



(S5 and S6 — the unsupported 18-ppg and Sweet 16 claims — are shown without relation edges: A neither implies nor contradicts them.)

8 Fifth example: a real legal case (*R v Renda*)

This example applies the task to a genuine judgment: *Renda*, *R v* [2005] EWCA Crim 2826 (England and Wales Court of Appeal, Criminal Division, 10 November 2005; also reported at [2006] 1 WLR 2948), supplied as the case file `clc/-ew-cases-EWCA-Crim-2005-2826.xml`. The judgment disposes of six conjoined appeals on the “bad character” provisions of the Criminal Justice Act 2003; we summarise the lead appeal, **Renda**, which contains several factual tensions that the relation analysis brings out.

Source (provenance)

Court of Appeal (Criminal Division), per the President of the Queen’s Bench Division (Sir Igor Judge). Appeal of Raymond Renda against conviction for attempted robbery before HHJ Van Der Werff and a jury at the Inner London Crown Court; the appeal turned on the Criminal Justice Act 2003 “false impression” (s 101(1)(f), s 105), “reprehensible behaviour” (s 112) and “contamination” (s 107) provisions.

Synthesized summary S

S states Renda’s self-serving evidence *first* and the conceded truths and the court’s holdings *afterward*, so that the later atoms positionally contradict the earlier ones, while prerequisite chains run from the charge through to the dismissal.

Raymond Renda was convicted of attempted robbery and appealed against that conviction. The charge arose when, at about 2 am on 10 November 2003, Renda accosted Robert Flint on Mile End Road, followed him home, and seized him by the neck demanding money; the trial turned on the relative credibility of Flint and Renda. To enhance

his credibility Renda testified to his good character: he claimed that his serious head injury had been sustained while he was serving on duty in the armed forces, and that he was currently in regular employment as a security guard. In fact, as he conceded under cross-examination, the head injury was sustained on holiday in a car accident, and his security work had been only short-term pass-checking so that he was no longer employed; he had therefore conveyed a false impression of positive good character. The defence argued that this cross-examination concession withdrew the false impression under s 105(3), but the court held that a concession forced in cross-examination is not a withdrawal. The court also recounted that in July 2001 Renda had been found unfit to plead to assault and given an absolute discharge, so that he stood not convicted, even though a jury had found as a fact that he struck a man from behind with a wooden table leg, which the court held to be reprehensible behaviour. Renda's counsel initially conceded that this table-leg finding amounted to a conviction, then later concluded that it was not a conviction. The court held that the evidence was not contaminated, and the appeal was dismissed.

Atomic units of S (self-contained, in order of appearance)

- L1.** Raymond Renda was convicted of attempted robbery.
- L2.** Raymond Renda appealed against the attempted-robbery conviction.
- L3.** Renda accosted Robert Flint on Mile End Road at about 2 am on 10 November 2003 and seized him by the neck while demanding money.
- L4.** The trial turned on the relative credibility of Robert Flint and Raymond Renda.
- L5.** Renda testified to his good character in order to enhance his credibility.
- L6.** Renda claimed his serious head injury was sustained while he was serving on duty in the armed forces.
- L7.** Renda claimed he was currently in regular employment as a security guard.
- L8.** In fact Renda's serious head injury was sustained on holiday in a car accident.
- L9.** In fact Renda's security work had been only short-term pass-checking and he was no longer employed.
- L10.** Renda had conveyed a false impression of positive good character.
- L11.** The defence argued that Renda's cross-examination concession withdrew the false impression under s 105(3).
- L12.** The court held that a concession forced in cross-examination is not a withdrawal of a false impression.
- L13.** In July 2001 Renda was found unfit to plead to assault and given an absolute discharge, so he stood not convicted.
- L14.** A jury found as a fact that Renda struck a man from behind with a wooden table leg.
- L15.** The court held that the table-leg incident was reprehensible behaviour.

L16. Renda’s counsel initially conceded that the table-leg finding amounted to a conviction.

L17. Renda’s counsel later concluded that the table-leg finding was not a conviction.

L18. The court held that the evidence was not contaminated and dismissed the appeal.

Relations

Logical prerequisites ($A \rightarrow B$).

- $L1 \rightarrow L2$ — a conviction is presupposed by an appeal against that conviction.
- $L3 \rightarrow L1$ — the attempted-robbery incident grounds the conviction for it.
- $L4 \rightarrow L5$ — the trial turning on credibility is why Renda testified to enhance his credibility.
- $L5 \rightarrow L6, L7$ — the good-character testimony comprises the two specific claims.
- $L8, L9 \rightarrow L10$ — the conceded truths are what establish that a false impression was conveyed.
- $L10 \rightarrow L11$ — the existence of a false impression is presupposed by arguing it was withdrawn.
- $L13, L14 \rightarrow L15$ — the unfit-to-plead finding plus the table-leg act ground the “reprehensible behaviour” holding.
- $L12, L15 \rightarrow L18$ — the holdings that defeated Renda’s challenges ground the dismissal of the appeal.

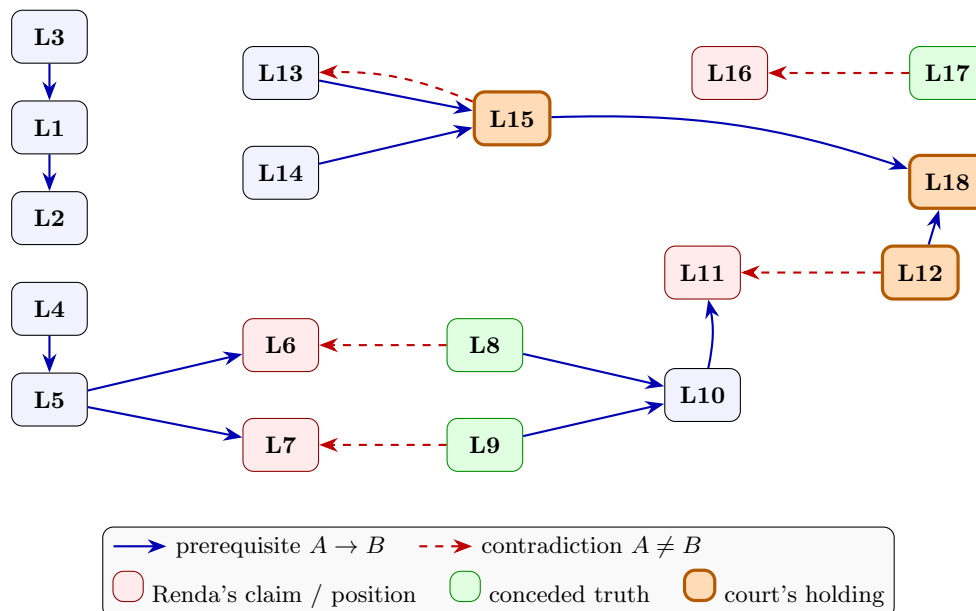
Contradictions / position invalidations ($A \neq B$). Each conceded truth or holding, stated later, contradicts an earlier self-serving claim or argument.

- $L8 \neq L6$ — injured on holiday in a car accident contradicts injured on duty in the armed forces.
- $L9 \neq L7$ — being no longer employed (short-term pass-checking) contradicts being currently in regular employment.
- $L12 \neq L11$ — the court’s holding that a forced concession is not a withdrawal defeats the defence’s s105(3) argument.
- $L15 \neq L13$ — “reprehensible behaviour” is held to stand despite there being no conviction, defeating the implied argument that no conviction means no reprehensible behaviour.
- $L17 \neq L16$ — counsel’s later conclusion (not a conviction) contradicts her initial concession (a conviction).

Note. The contradictions cluster on the credibility issue (Renda’s claims L6/L7 vs. the conceded truths L8/L9) and on two reversed legal positions (L11/L12 and L16/L17), plus the not-convicted-vs.-reprehensible tension (L13/L15). The prerequisite spine runs $L3 \rightarrow L1 \rightarrow L2$ and, via the false-impression and bad-character findings, to the dismissal L18.

Relation graph

Solid blue arrows are logical prerequisites ($A \rightarrow B$); dashed red arrows are contradictions ($A \neq B$). Renda’s self-serving claims/positions are tinted red; the conceded truths and the court’s holdings that override them are green; the dismissal L18 is highlighted. Nodes are placed on an explicit grid so no edge crosses a node.



(For legibility, purely sequential narration links among the spine atoms L2–L5 and the bracketing of L16/L17 within the contamination argument are summarised rather than drawn edge-by-edge.)

A faithful, naturally-ordered summary K

The summary S above was deliberately ordered (claims first, conceded truths afterward) to manufacture positional contradictions. For contrast we now give a *faithful* summary **K** of the same Renda appeal that follows the judgment’s own logical and chronological order: each disputed claim is reported *together with* its resolution, and the court’s reasoning is presented in the sequence the court itself adopts. Under this ordering the relation structure is almost entirely *logical implication*; the only residual tensions are the two points the court expressly adjudicates, and even these read as holdings that *resolve* an issue rather than as a later atom flatly contradicting an earlier one.

Raymond Renda was convicted of attempted robbery before HHJ Van Der Werff and a jury, and appealed. The charge arose when, at about 2 am on 10 November 2003, he accosted Robert Flint on Mile End Road, followed him home, and seized him by the neck while demanding money; because the only real question was whether any offence had been committed, the trial turned on the relative credibility of Flint and Renda. Seeking to present himself as a man of positive good character, Renda testified that his serious head injury had been sustained on duty in the armed forces and that he was in regular employment as a security guard; in truth, however, the injury was sustained on holiday in a car accident and the security work had been only short-term, so that this evidence conveyed a false impression within s 105(1). When Renda conceded these matters under cross-examination, the court held that a concession extracted in cross-examination does

not amount to a withdrawal of the false impression under s 105(3). To help correct that false impression, the Crown was permitted to put a July 2001 incident before the jury: Renda had then been found unfit to plead to assault and given an absolute discharge, so he was not convicted, yet a jury had found as a fact that he struck a man from behind with a wooden table leg, and the court held that this act was reprehensible behaviour within the bad-character provisions notwithstanding the absence of a conviction. A submission that the resulting evidence was contaminated under s 107 was rejected, the court holding that it was not contaminated, and the appeal was dismissed.

Atomic units of K (self-contained, in order of appearance)

- K1.** Raymond Renda was convicted of attempted robbery and appealed against the conviction.
- K2.** Renda accosted Robert Flint on Mile End Road at about 2 am on 10 November 2003, followed him home, and seized him by the neck while demanding money.
- K3.** The trial turned on the relative credibility of Robert Flint and Raymond Renda.
- K4.** Renda testified seeking to present himself as a man of positive good character.
- K5.** Renda testified that his head injury was sustained on duty in the armed forces and that he was in regular employment as a security guard.
- K6.** In truth Renda's injury was sustained on holiday in a car accident and his security work had been only short-term, so his evidence conveyed a false impression within s 105(1).
- K7.** Renda conceded these matters under cross-examination.
- K8.** The court held that a concession extracted in cross-examination does not amount to a withdrawal of the false impression under s 105(3).
- K9.** To help correct the false impression, the Crown was permitted to put a July 2001 incident before the jury.
- K10.** In the July 2001 incident Renda was found unfit to plead to assault and given an absolute discharge, so he was not convicted.
- K11.** A jury had found as a fact that Renda struck a man from behind with a wooden table leg.
- K12.** The court held that the table-leg act was reprehensible behaviour within the bad-character provisions, notwithstanding the absence of a conviction.
- K13.** A submission was made that the evidence was contaminated under s 107.
- K14.** The court held that the evidence was not contaminated.
- K15.** The appeal was dismissed.

Relations

Logical prerequisites ($A \rightarrow B$).

- $K2 \rightarrow K1$ — the attempted-robbery incident grounds the conviction and the appeal.
- $K1 \rightarrow K3$ — the contested conviction is what makes credibility the trial issue.
- $K3 \rightarrow K4$ — credibility being decisive is why Renda testified to his good character.
- $K4 \rightarrow K5$ — the good-character case comprises the specific claims about injury and employment.
- $K5 \rightarrow K6$ — the claims are what the truth contradicts, yielding the “false impression” finding.
- $K6 \rightarrow K7$ — there must be a false assertion for Renda to concede it under cross-examination.
- $K7 \rightarrow K8$ — the cross-examination concession is the act the s 105(3) holding rules upon.
- $K6 \rightarrow K9$ — the false impression is what the July 2001 material is admitted to correct.
- $K10, K11 \rightarrow K12$ — the unfit-to-plead finding and the found-fact assault ground the “reprehensible behaviour” holding.
- $K9 \rightarrow K13$ — admitting the bad-character material is the premise of the contamination submission.
- $K13 \rightarrow K14$ — the contamination submission is what the “not contaminated” holding answers.
- $K8, K12, K14 \rightarrow K15$ — the three holdings against Renda jointly ground the dismissal.

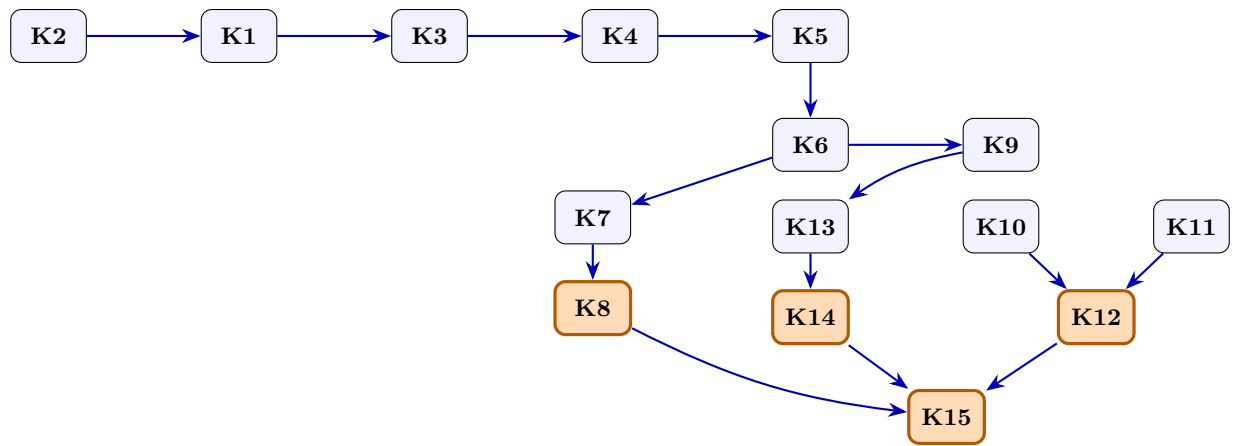
Contradictions / position invalidations ($A \neq B$). Under the faithful ordering there are *no* positional invalidations between asserted atoms: each claim (K5) is reported together with its correction (K6), so the truth does not appear later as a contradicting atom but as part of the same finding. The two genuine tensions in the case are stated as *holdings that resolve* an issue, not as contradictions:

- K8 settles the s 105(3) question (a forced concession is not a withdrawal) — presented as a ruling, with no competing asserted atom left standing.
- K12 settles the reprehensible-behaviour question (conduct can be reprehensible though it produced no conviction, K10) — again a ruling rather than a flat contradiction.

Note. Comparing K with S (the same facts, deliberately reordered) illustrates the point: positional invalidation is a property of how a text is *arranged*, not only of the facts it contains. A faithful, logically ordered summary of *Renda* carries the case’s full content as an implication chain from the incident (K2) to the dismissal (K15), with the disputed points absorbed into the findings rather than surfacing as contradictions.

Relation graph

Solid blue arrows are logical prerequisites ($A \rightarrow B$). The court’s three operative holdings (K8, K12, K14) are highlighted; together they ground the dismissal K15. There are no contradiction edges — the faithful ordering folds each disputed claim into its finding. Nodes are placed on an explicit grid so no edge crosses a node.



→ prerequisite $A \rightarrow B$ court's operative holding
 No contradiction edges: faithful ordering folds each disputed claim into its finding.